

THE EIGHTH WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION

March 20, 1948

Morning Session

1. What is the maximum of $|z^3 - z + 2|$; where z is a complex number with $|z| = 1$? (page 243)

2. Two spheres in contact have a common tangent cone. These three surfaces divide the space into various parts, only one of which is bounded by all three surfaces; it is "ring-shaped." Being given the radii of the spheres, r and R , find the volume of the "ring-shaped" part. (The desired expression is a rational function of r and R .) (page 243)

3. Let $\{a_n\}$ be a decreasing sequence of positive numbers with limit 0 such that

$$b_n = a_n - 2a_{n+1} + a_{n+2} \geq 0$$

for all n . Prove that

$$\sum_{n=1}^{\infty} nb_n = a_1. \quad \text{(page 247)}$$

4. Let D be a plane region bounded by a circle of radius r . Let (x, y) be a point of D and consider a circle of radius δ and center at (x, y) . Denote by $l(x, y)$ the length of that arc of the circle which is outside D . Find

$$\lim_{\delta \rightarrow 0} \frac{1}{\delta^2} \int \int_D l(x, y) \, dx \, dy. \quad \text{(page 248)}$$

Note. This is the text of the problem as published in the *American Mathematical Monthly*, vol. 55 (1948) p. 632. Clearly the " d " under the limit sign should be a " δ ." We do not know how the problem was printed on the actual examination.

5. If $x_1, \dots, x_n$ denote the n th roots of unity, evaluate

$$\pi(x_i - x_j)^2 \quad (i < j). \quad \text{(page 249)}$$

Note. This is the text of the problem as published in the *American Mathematical Monthly*, vol. 55 (1948) p. 632. The “ π ” should be a product sign. Again, we do not know how the problem was printed on the actual examination.

6. Answer either (i) or (ii):

(i) A force acts on the element ds of a closed plane curve. The magnitude of this force is $r^{-1} ds$ where r is the radius of curvature at the point considered, and the direction of the force is perpendicular to the curve; it points to the convex side. Show that the system of such forces acting on all elements of the curve keep it in equilibrium. (page 252)

(ii) Show that

$$x + \frac{2}{3}x^3 + \frac{2}{3}\frac{4}{5}x^5 + \frac{2}{3}\frac{4}{5}\frac{6}{7}x^7 + \dots = \frac{\arcsin x}{\sqrt{1-x^2}}.$$

(page 254)

Afternoon Session

1. Let $f(x)$ be a cubic polynomial with roots x_1, x_2 , and x_3 . Assume that $f(2x)$ is divisible by $f'(x)$ and compute the ratios $x_1:x_2:x_3$. (page 255)

2. “A penny in a corner.” A circle moves so that it is continually in contact with all three coordinate planes of an ordinary rectangular system. Find the locus of the center of the circle. (page 256)

3. If n is a positive integer, prove that

$$[\sqrt{n} + \sqrt{n+1}] = [\sqrt{4n+2}],$$

where $[x]$ denotes as usual the greatest integer not exceeding x . (page 257)

4. Let $\min(x, y)$ denote the smaller of the numbers x and y . For what λ 's does the equation

$$\int_0^1 \min(x, y)f(y)dy = \lambda f(x)$$

have continuous solutions which do not vanish identically in $(0, 1)$? What are these solutions? (page 258)

5. The pairs of numbers (a, b) such that $|a + bt + t^2| \leq 1$ for $0 \leq t \leq 1$ fill a certain region in the (a, b) -plane. What is the area of this region? (page 259)

6. Answer either (i) or (ii):

(i) Let V_1, V_2, V_3 , and V denote four vertices of a cube. V_1, V_2 , and V_3 are next neighbors of V , that is, the lines VV_1, VV_2 , and VV_3 are edges of

the cube. Project the cube orthogonally onto a plane (the z -plane, the Gaussian plane) of which the points are marked with complex numbers. Let the projection of V fall in the origin and the projections of V_1 , V_2 , and V_3 in points marked with the complex numbers z_1 , z_2 , and z_3 , respectively. Show that $z_1^2 + z_2^2 + z_3^2 = 0$. (page 261)

(ii) Let a_{ij} be a determinant in which each diagonal element exceeds in absolute value the sum of the absolute values of the other elements of its row, that is

$$|a_{ii}| > |a_{i1}| + |a_{i2}| + \cdots + |a_{i,i-1}| + |a_{i,i+1}| + \cdots + |a_{in}|.$$

Show that the determinant is not equal to zero. (Consider the corresponding system of linear homogeneous equations.) (page 263)